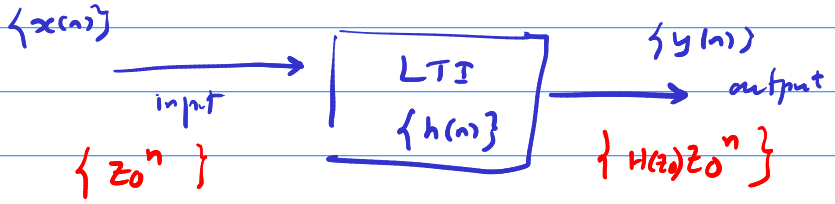
Z - Transform

- Analytical tool to study or design LTI systems

Eigen function/signal for LTI systems



Input: General complex exponential

$x(n) = z_0^n \; ; \; n \in Z$

z_0 is a complex number

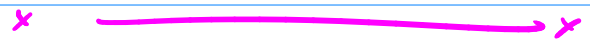
$$y(n) = \sum_{k=-\infty}^{\infty} h(k)\, x(n-k)$$

$x(n-k) = z_0^{n-k}$

$$= z_0^n \sum_{k=-\infty}^{\infty} h(k)\, z_0^{-k}$$

$\sum_{k=-\infty}^{\infty} h(k)\, z_0^{-k} \Rightarrow H(z_0)$ complex gain

$y(n) = H(z_0)\, z_0^n \; ; \; n \in Z$



Definition: z-transform of DT signal $\{x(n)\}$ is defined as

$$(*) \qquad X(z) = \sum_{n=-\infty}^{\infty} x(n)\, z^{-n}$$

$z \rightarrow$ complex variable.

$$X(z) : \mathbb{C} \longrightarrow \mathbb{C}$$

mapping $z \longrightarrow X(z)$

$X(z) \Rightarrow$ complex valued function defined over a complex variable.

Region of Convergence: (ROC)

Set of all values of z for which $(*)$ absolutely converges is called ROC.

$$\sum_{n=-\infty}^{\infty} |x(n)\, z^{-n}| < \infty$$

Examples:

1. $\{x(n)\} = \{\delta(n)\}$

$$= \{\ldots, 0, \underset{\uparrow}{1}, 0, \ldots\} \quad (n=0)$$

$$X(z) = 1 \cdot z^0$$

$$= 1 \quad \forall\ z \in \mathbb{C}$$

ROC = Entire complex plane (z)

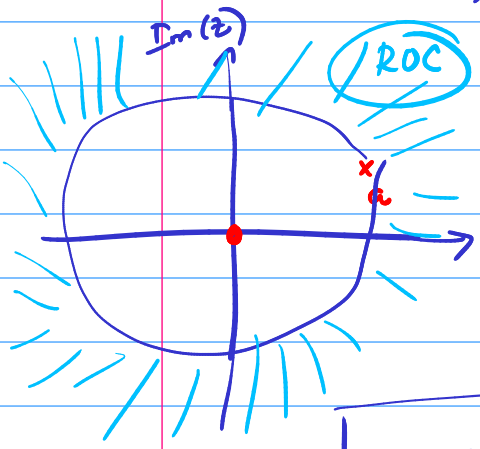
2. Causal Exponential

$$x(n) = a^n u(n) \qquad a \in \mathbb{C}$$

$$= \{\underset{\uparrow}{1}, a, a^2, \ldots\} \quad (n=0)$$

$\times \rightarrow$ Pole

$\bullet \rightarrow$ zero



$$X(z) = \sum_{n=0}^{\infty} a^n z^{-n}$$

$$= \sum_{n=0}^{\infty} (az^{-1})^n$$

Geometric series

$$\sum_{k=0}^{\infty} r^k = \frac{1}{1-r} \quad \text{if } |r| < 1$$

$$= \frac{1}{1 - az^{-1}} \quad ; \quad |az^{-1}| < 1$$

$$\frac{|a|}{|z|} < 1$$

$$X(z) = \frac{1}{1 - az^{-1}} \quad ; \quad \underbrace{|z| > |a|}_{\text{ROC}}$$

$$= \frac{z}{z-a}$$

3. Anti - causal Exponential

$$x(n) = -a^n u(-n-1)$$

$$= \{\ldots, -a^{-2}, -a^{-1}, \underset{n=0}{\underset{\downarrow}{0}}, \ldots\}$$

$$X(z) = \sum_{n=-1}^{-\infty} -a^n z^{-n}$$

$$k = -n$$

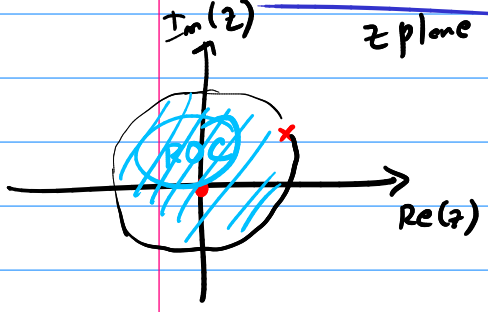
$$= +1 - 1 \sum_{k=1}^{\infty} -a^{-k} z^k$$

$$= 1 - \sum_{k=0}^{\infty} (a^{-1}z)^k$$

$$X(z) = 1 - \frac{1}{1 - a^{-1}z} \quad ; \quad |a^{-1}z| < 1$$

$$\frac{|z|}{|a|} < 1$$

$$\boxed{X(z) = \frac{1}{1 - az^{-1}} \quad ; \quad \underbrace{|z| < |a|}_{\text{ROC}}}$$



$$= \frac{z}{z - a} \quad ; \quad |z| < |a|$$

$X(z)$ => Rational z-transform if

$$X(z) = \frac{A(z)}{B(z)}$$

$B(z)$, $A(z)$ => Polynomials

- Finite length signals

  $n=0$

  causal $\{x(0), \ldots, x(N)\}$

$$X(z) = x(0)z^{0} + x(1)z^{-1} + \ldots + x(N)z^{-N}$$

Poles at $z=0$ (N^{th} order) $= \dfrac{x(0)z^{N} + \ldots + x(N)}{z^{N}}$

- anticausal $\{x(-N), \ldots, x(0)\}$ ($n=0$)

Pole at $z=\infty$ $X(z) = x(-N)z^{N} + \ldots + x(-1)z + x(0)$

- Non causal (two sided) ($n=0$)

Pole at $z=0$ both $z=\infty$ $\{x(-N), \ldots, x(0), \ldots, x(N)\}$

$$X(z) = x(-N)z^{N} + \ldots + x(0) + \ldots + x(N)z^{-N}$$

$X(z) = \dfrac{A(z)}{B(z)}$

$X(z)$: Poles : Roots of Denominator $B(z)$

Zeros : Roots of Numerator $A(z)$

$X(z) = 0$ if z is root of $A(z)$ (zero)

$|X(z)| = \infty$ if z is root of $B(z)$ (Pole)

Poles or zeros at (origin) $z=0$ are called trivial

$|X(z)| \to \infty$ when $z \to$ pole

- Two sided exponential

$$x(n) = a^n u(n) - b^n u(-n-1), \quad n \in \mathbb{Z}$$

(causal) (anticausal)

(right sided) (left sided)

$$\sum x(n) z^{-n}$$

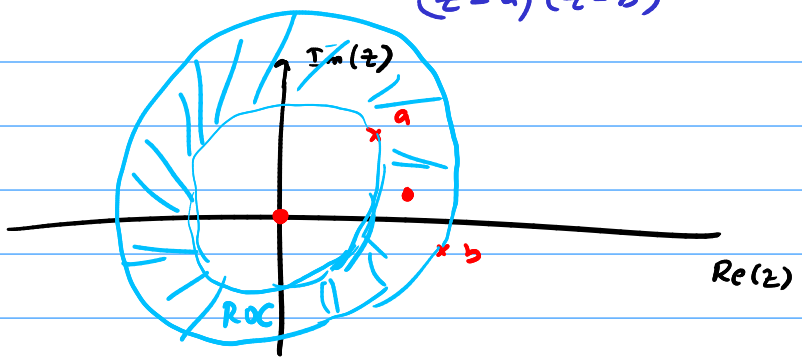
$$X(z) = \frac{1}{1-az^{-1}} + \frac{1}{1-bz^{-1}}$$

$ROC_1: |z| > |a|$ $\quad$ $ROC_2: |z| < |b|$

ROC of $X(z)$: $ROC_1 \cap ROC_2$

$|a| < |b|$

$$X(z) = \frac{z(2z - (a+b))}{(z-a)(z-b)}$$; ROC $|a| < |z| < |b|$



Poles @ a, b

zeros @ 0, $\frac{a+b}{2}$

If $|b| < |a|$ then $X(z)$ doesn't exist

ROC = { }

empty set

Properties of ROC

$$X(z) = \sum_{n=-\infty}^{\infty} x(n)\, z^{-n}$$

Requirement — Absolute Summability

$$\sum_n |x(n) z^{-n}| < \infty$$

$z = re^{j\omega}$ $\quad r = |z|$ Magnitude

$\omega = \angle z$ Phase

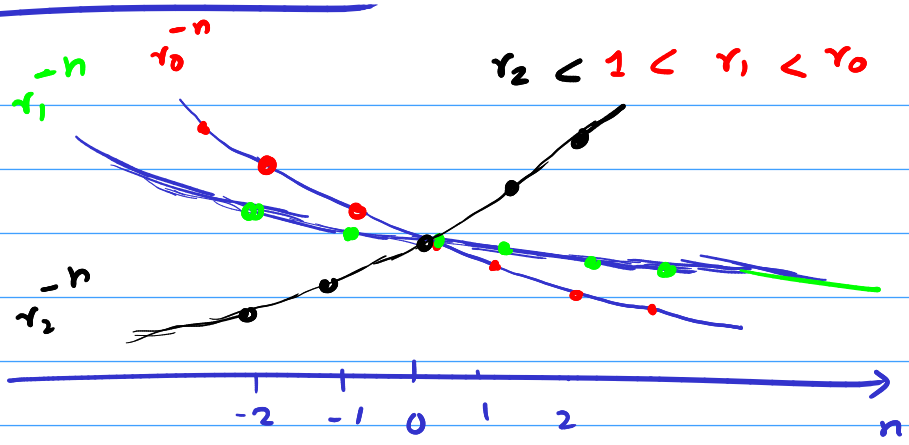
$$\sum_n |x(n)|\, |re^{j\omega}|^{-n} < \infty$$

$|e^{j\omega}| = 1$

$$\sum_n |x(n)|\, r^{-n} < \infty$$

$|z| \Rightarrow$ governs convergence of $X(z)$

Behavior of r^{-n} $(r > 0)$



$x(n) \Rightarrow$ Two sided signal

Absolute convergence of $X(z)$

$$\sum_{n=0}^{\infty} |x(n)| r^{-n} \quad \text{(causal)} \qquad + \qquad \sum_{n=-1}^{-\infty} |x(n)| r^{-n} \quad \text{(anti causal)}$$

If causal part converges for $r = r_c$ then it will also converge for any $r > r_c$

If anti causal part converges for $r = r_a$ then it will also converge for any $r < r_a$

ROC will be in general of form

$$r_c < |z| < r_a$$

Some cases. r_c can be 0 r_a can be ∞

ROC Properties:

1. ROC is governed by only $|z|$

   Phase of z does not affect ROC.

2. ROC is always a connected region.

3. Finite length signals have entire

   z-plane as ROC

   except possibly $z=0$ and/or $z=\infty$

4. Rational z-transform $X(z) = \frac{A(z)}{B(z)}$ (A: polynomial, B: polynomial)

   Roots of $A(z)$ $\Rightarrow$ zeros

   Roots of $B(z)$ $\Rightarrow$ poles.

ROC is governed by poles

Zeros do not affect ROC.

- For causal signal (right sided)

  $(a^n u(n) + b^n u(n) + c^n u(n))$

  Poles @ a, b, c

  ROC will be outside lowest pole

- For anticausal signal (left sided)

  ROC will lie inside smallest pole.

- Two sided Signal

  ROC will be in between two poles.

$x(n) = 2^n u(n) + 3^n u(n) - 4^n u(-n-1)$

$|z| > |2|$ $\quad$ Poles @ 2, 3, 4 $\quad$ $|z| < |4|$

$|z| > |3|$

Roc $3 < |z| < 4$

$x(n) = 2^n u(n) - 3^n u(-n-1) - 4^n u(-n-1)$

Poles @ 2, 3, 4

Roc $2 < |z| < 3$